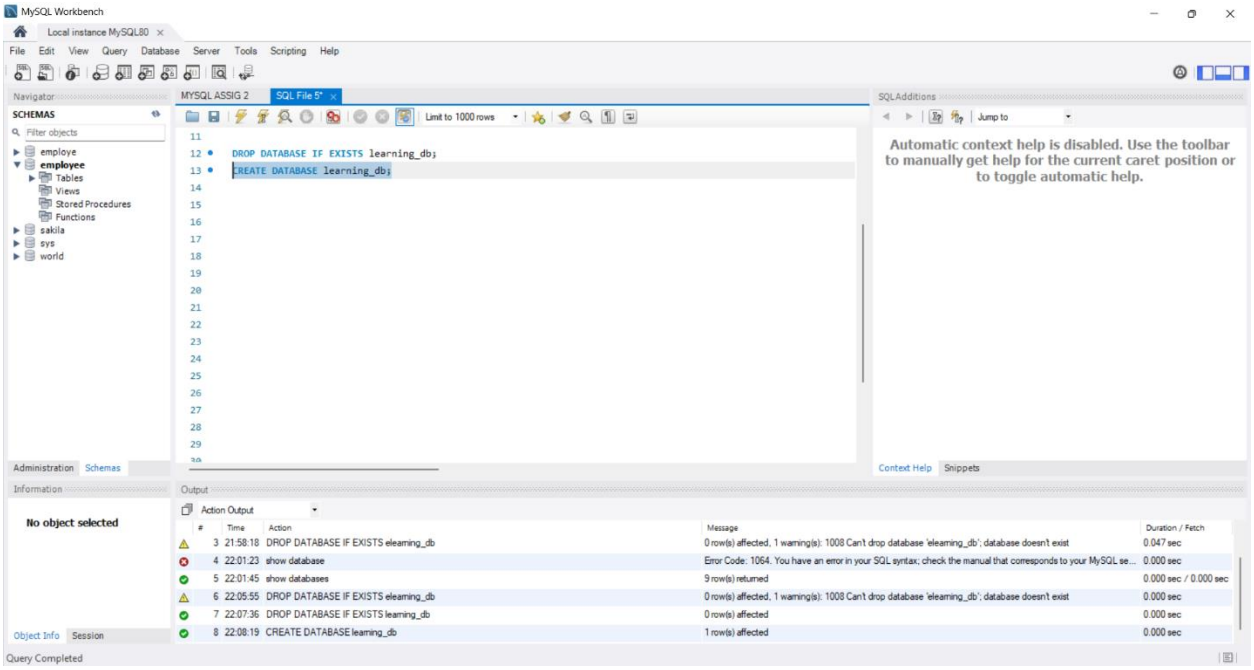
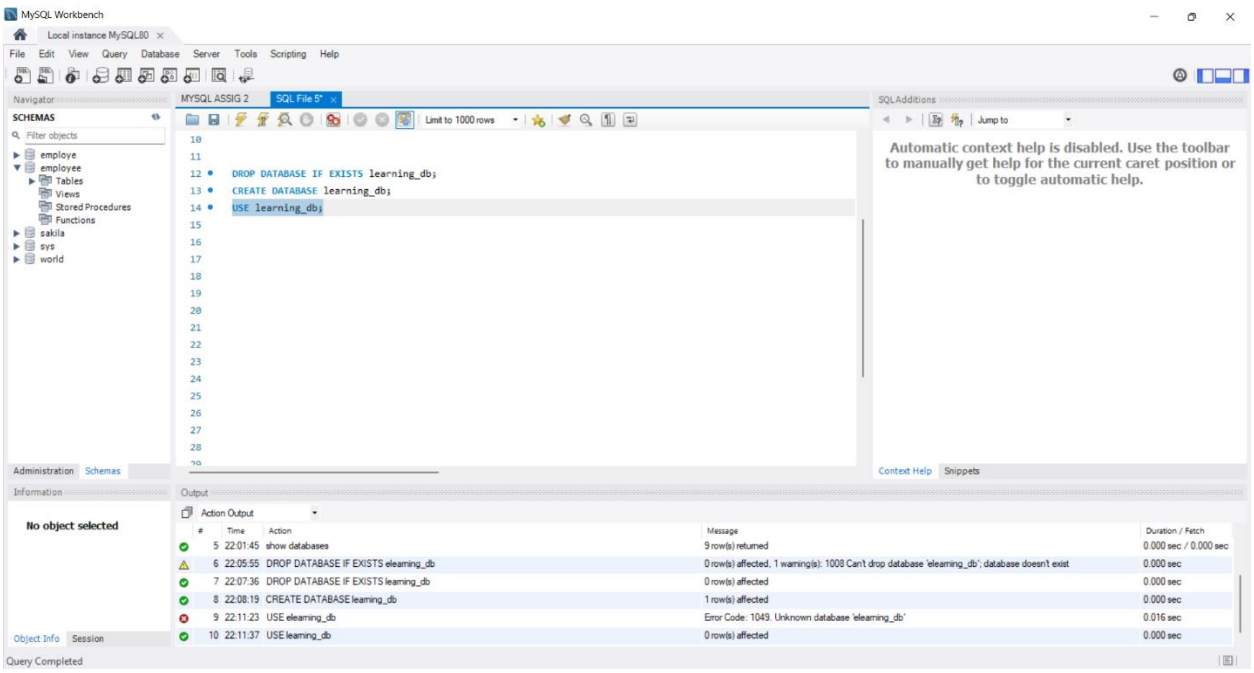
MODULE END ASSIGNMENT

ANALYZING E-LEARNING PLATFORM PURCHASES USING MYSQL

1. Created the database and schemas.

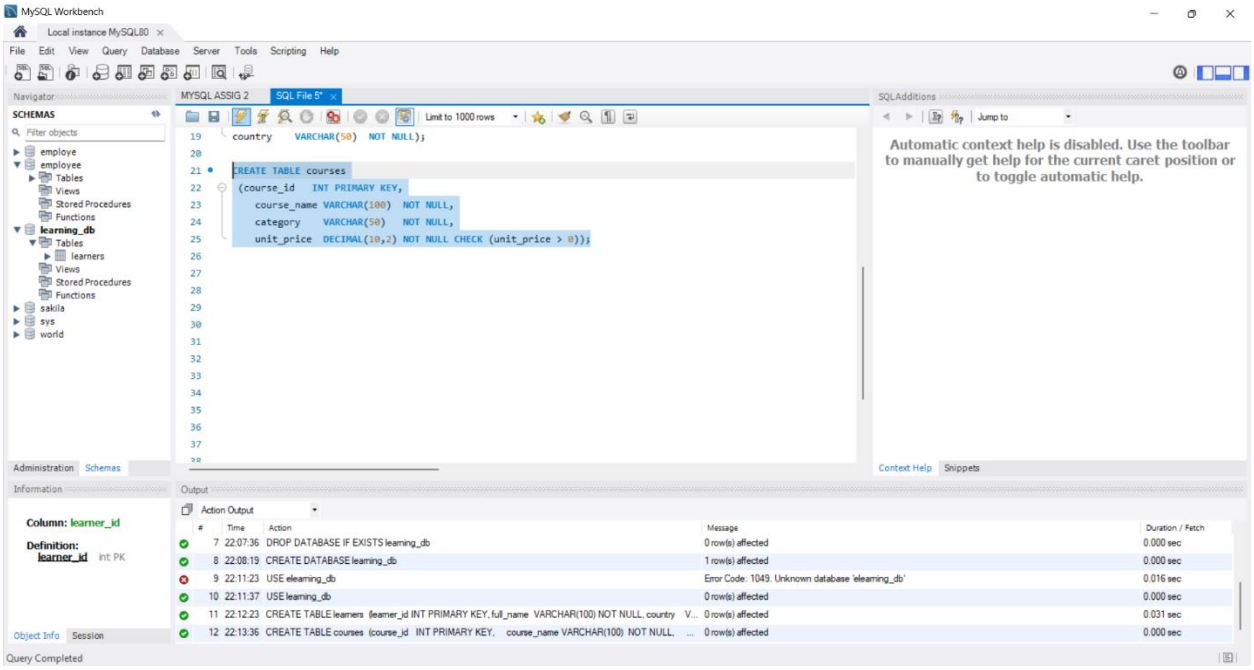
Created a Database for this project

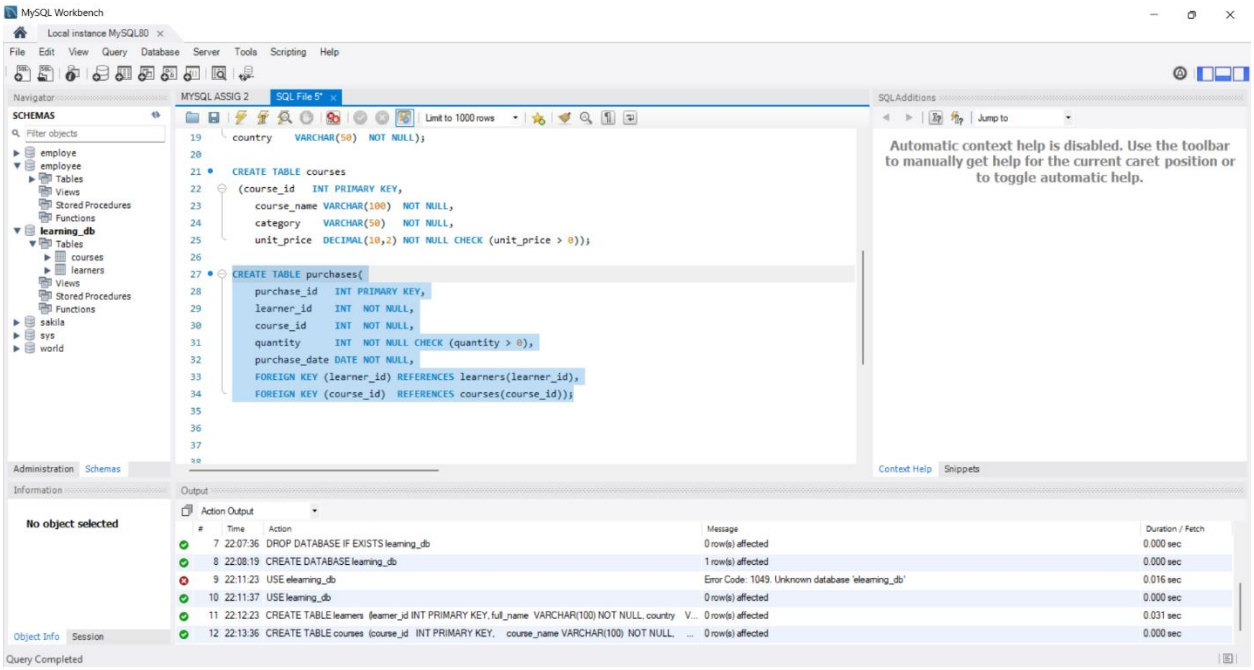




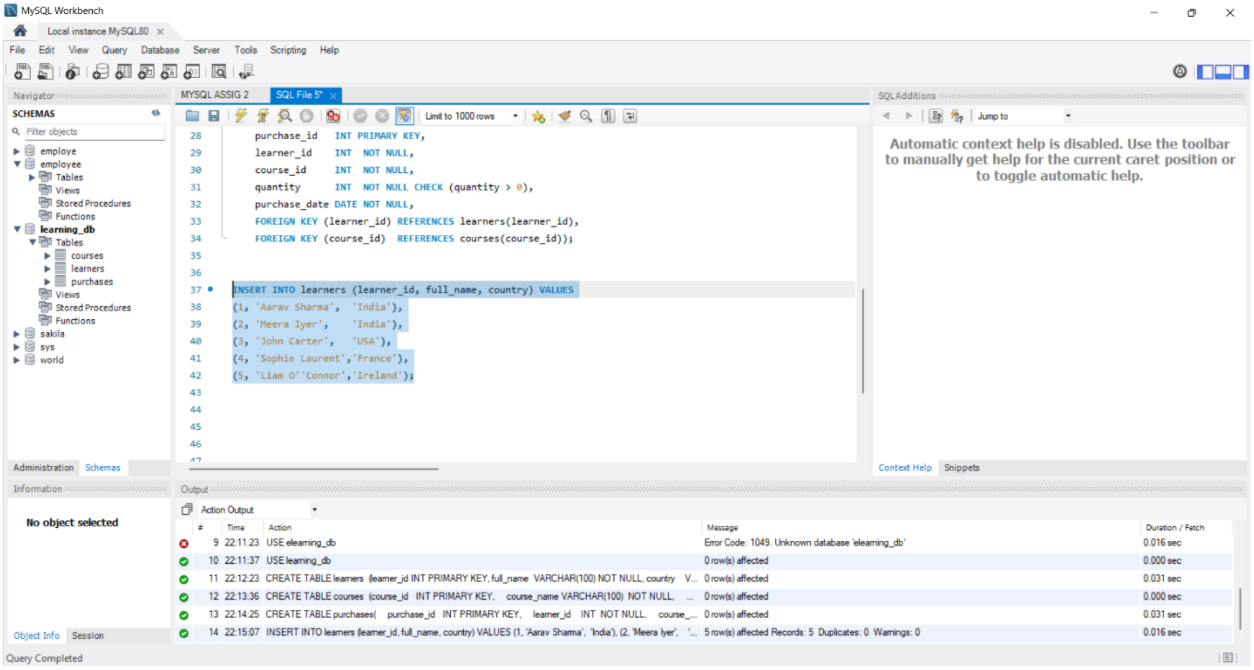
2.Create the Three Tables

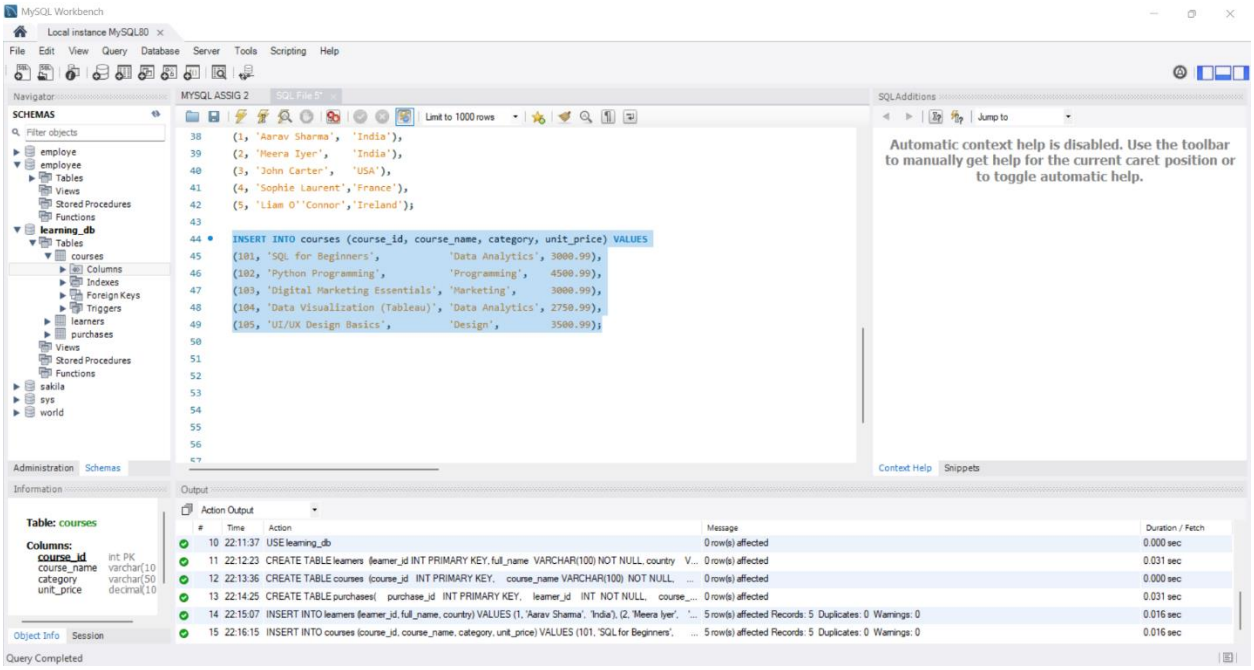
Create all three tables in MySQL with appropriate **data types** and **relationships**.

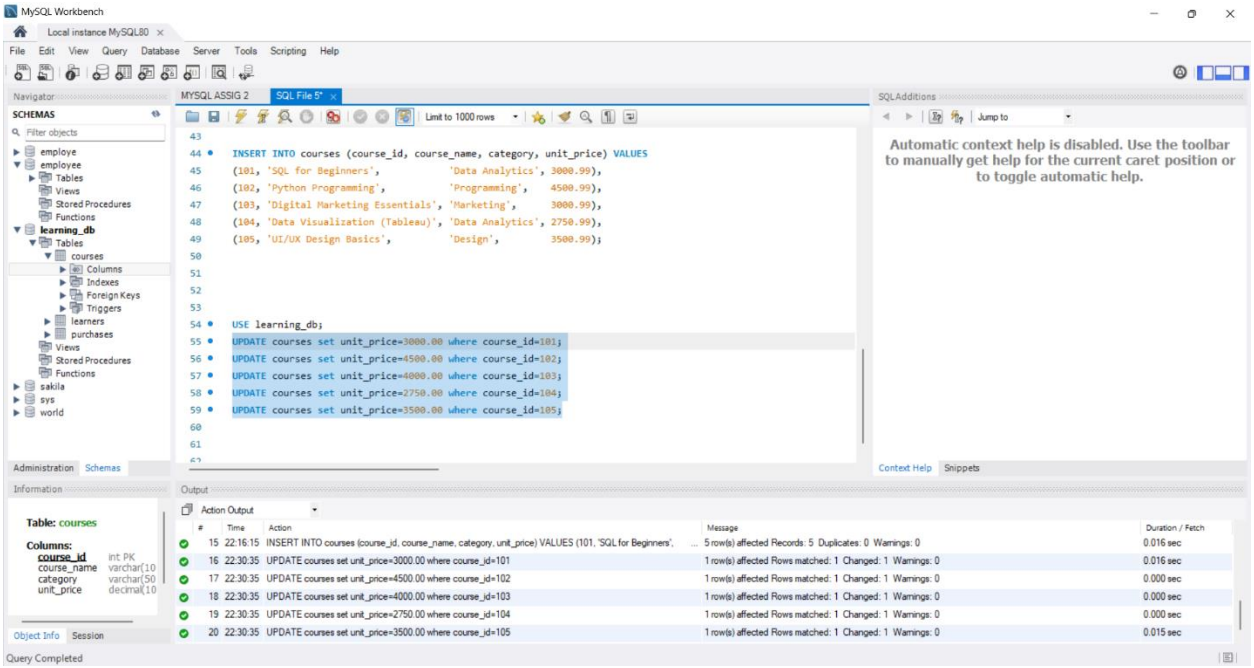

MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
employe
employee
Tables
Views
Stored Procedures
Functions
learning_db
Tables
learners
Views
Stored Procedures
Functions
sakila
sys
world
Administration Schemas
Information
Column: learner_id
Definition:
learner_id int PK
Object Info Session
MYSQL ASSIG 2
SQL File 5*
Limit to 1000 rows
19 country VARCHAR(50) NOT NULL);
20
21 CREATE TABLE courses
22 (course_id INT PRIMARY KEY,
23 course_name VARCHAR(100) NOT NULL,
24 category VARCHAR(50) NOT NULL,
25 unit_price DECIMAL(10,2) NOT NULL CHECK (unit_price > 0));
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
7 22:07:36 DROP DATABASE IF EXISTS learning_db 0 row(s) affected 0.000 sec
8 22:08:19 CREATE DATABASE learning_db 1 row(s) affected 0.000 sec
9 22:11:23 USE elearning_db Error Code: 1049. Unknown database 'elearning_db' 0.016 sec
10 22:11:37 USE learning_db 0 row(s) affected 0.000 sec
11 22:12:23 CREATE TABLE learners (learner_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, country V... 0 row(s) affected 0.031 sec
12 22:13:36 CREATE TABLE courses (course_id INT PRIMARY KEY, course_name VARCHAR(100) NOT NULL, ... 0 row(s) affected 0.000 sec
Query Completed

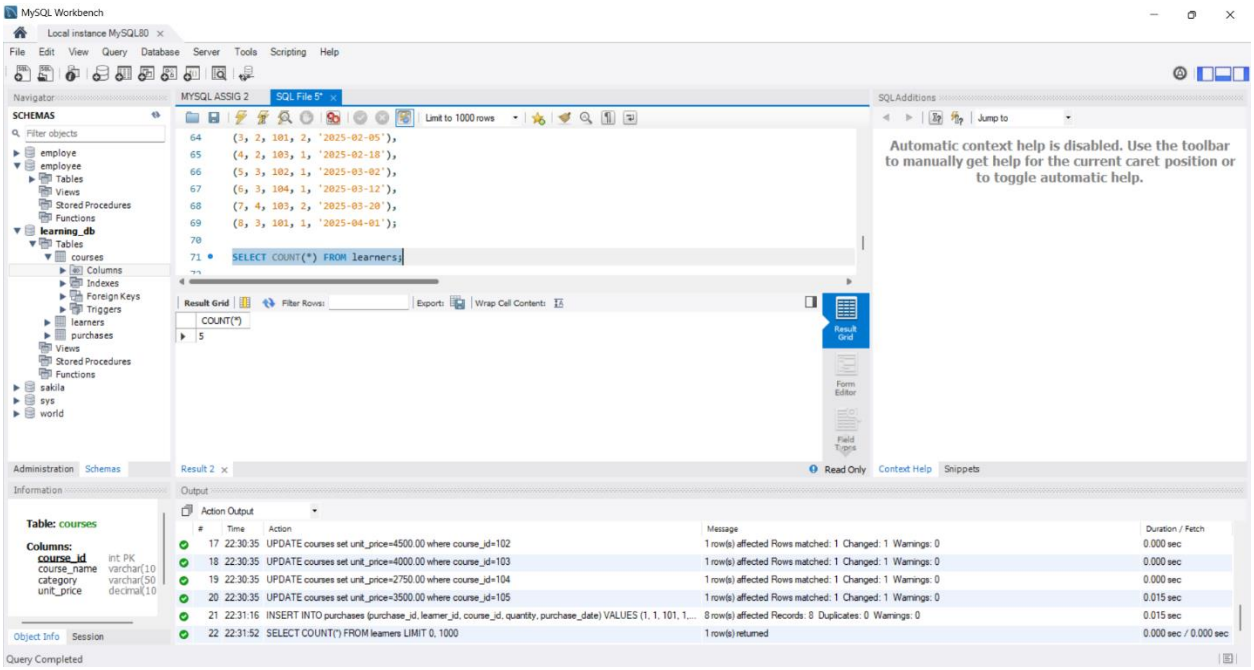

MySQL Workbench
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Stored Procedures
Functions
sakila
sys
world
Administration Schemas
Information
No object selected
Object Info Session
MYSQL ASSIG 2
SQL File 5*
Limit to 1000 rows
19 country VARCHAR(50) NOT NULL);
20
21 CREATE TABLE courses
22 (course_id INT PRIMARY KEY,
23 course_name VARCHAR(100) NOT NULL,
24 category VARCHAR(50) NOT NULL,
25 unit_price DECIMAL(10,2) NOT NULL CHECK (unit_price > 0));
26
27 CREATE TABLE purchases(
28 purchase_id INT PRIMARY KEY,
29 learner_id INT NOT NULL,
30 course_id INT NOT NULL,
31 quantity INT NOT NULL CHECK (quantity > 0),
32 purchase_date DATE NOT NULL,
33 FOREIGN KEY (learner_id) REFERENCES learners(learner_id),
34 FOREIGN KEY (course_id) REFERENCES courses(course_id));
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
7 22:07:36 DROP DATABASE IF EXISTS learning_db 0 row(s) affected 0.000 sec
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Query Completed

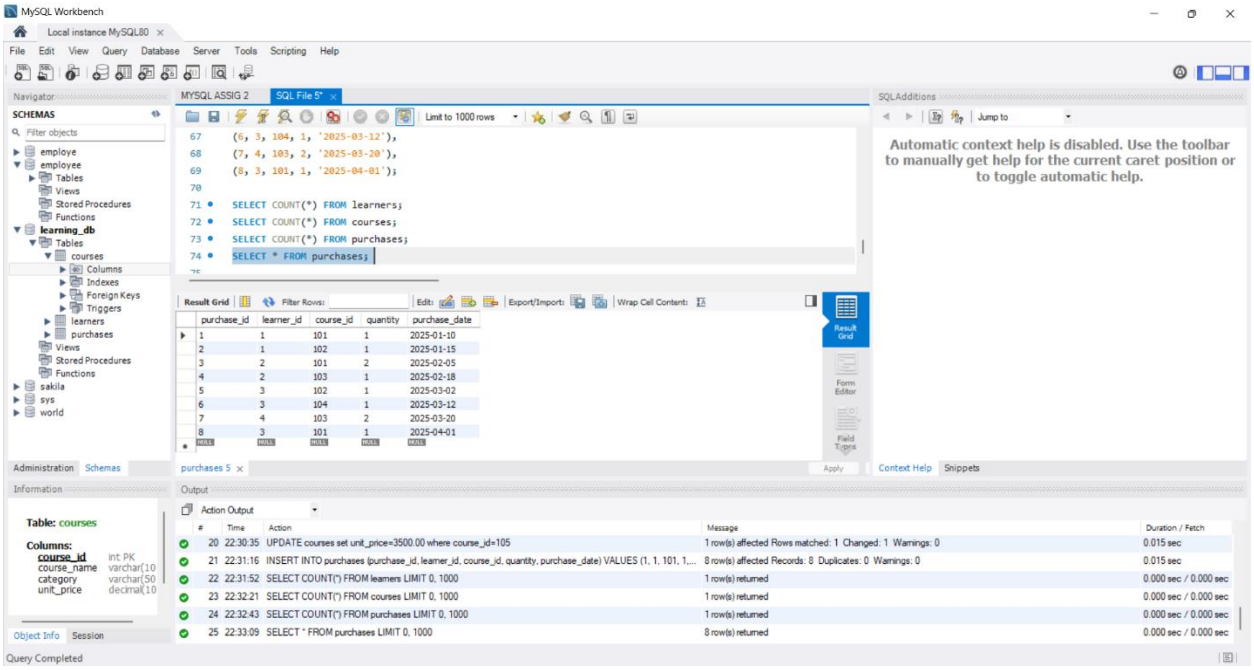
3: Inserted sample data





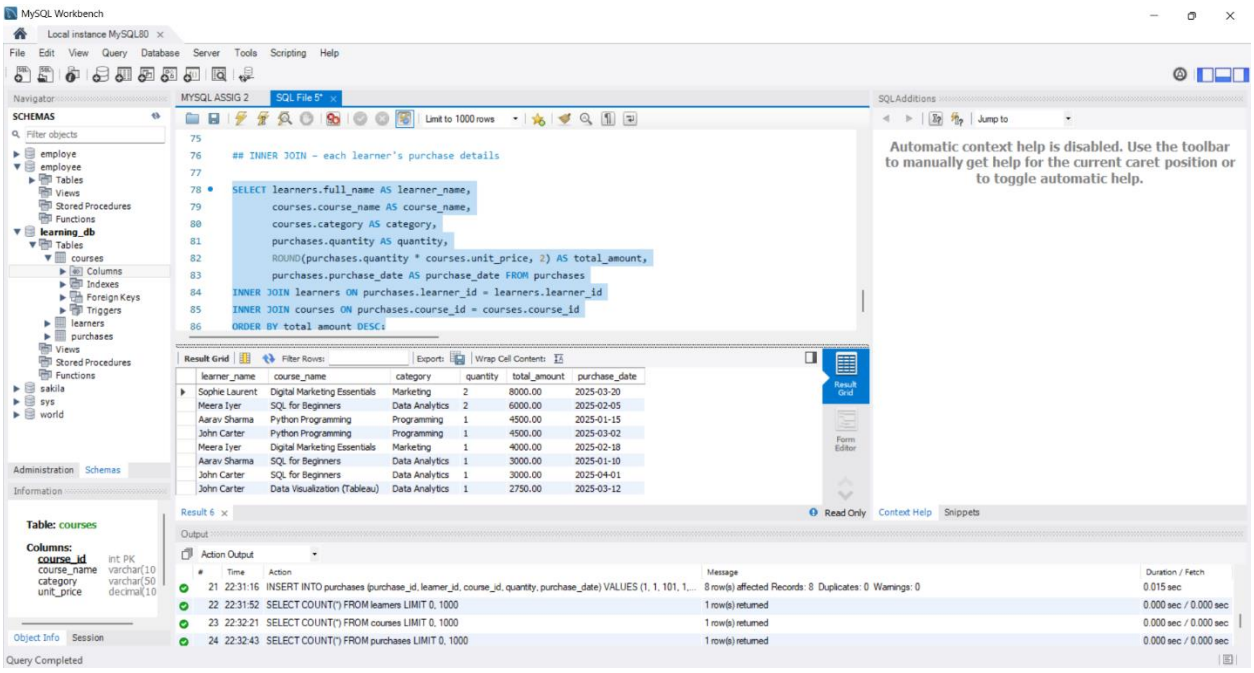

MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
MYSQL ASSIG 2
SQL File 5*
Limit to 1000 rows
INSERT INTO courses (course_id, course_name, category, unit_price) VALUES
(101, 'SQL for Beginners', 'Data Analytics', 3000.99),
(102, 'Python Programming', 'Programming', 4500.99),
(103, 'Digital Marketing Essentials', 'Marketing', 3000.99),
(104, 'Data Visualization (Tableau)', 'Data Analytics', 2750.99),
(105, 'UI/UX Design Basics', 'Design', 3500.99);
USE learning_db;
UPDATE courses set unit_price=3000.00 where course_id=101;
UPDATE courses set unit_price=4500.00 where course_id=102;
UPDATE courses set unit_price=4000.00 where course_id=103;
UPDATE courses set unit_price=2750.00 where course_id=104;
UPDATE courses set unit_price=3500.00 where course_id=105;
SQLAdditions
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Output
Action Output
15 22:16:15 INSERT INTO courses (course_id, course_name, category, unit_price) VALUES (101, 'SQL for Beginners', ... 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 0.016 sec
16 22:30:35 UPDATE courses set unit_price=3000.00 where course_id=101 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 0.016 sec
17 22:30:35 UPDATE courses set unit_price=4500.00 where course_id=102 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 0.000 sec
18 22:30:35 UPDATE courses set unit_price=4000.00 where course_id=103 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 0.000 sec
19 22:30:35 UPDATE courses set unit_price=2750.00 where course_id=104 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 0.000 sec
20 22:30:35 UPDATE courses set unit_price=3500.00 where course_id=105 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 0.015 sec
Table: courses
Columns:
course_id int PK
course_name varchar(10
category varchar(50
unit_price decimal(10
Query Completed


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
MYSQL ASSIG 2
SQL File 5*
Limit to 1000 rows
(3, 2, 101, 2, '2025-02-05'),
(4, 2, 103, 1, '2025-02-18'),
(5, 3, 102, 1, '2025-03-02'),
(6, 3, 104, 1, '2025-03-12'),
(7, 4, 103, 2, '2025-03-20'),
(8, 3, 101, 1, '2025-04-01');
SELECT COUNT(*) FROM learners;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
COUNT(*)
5
Result 2
Read Only
SQLAdditions
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Output
Action Output
17 22:30:35 UPDATE courses set unit_price=4500.00 where course_id=102 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 0.000 sec
18 22:30:35 UPDATE courses set unit_price=4000.00 where course_id=103 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 0.000 sec
19 22:30:35 UPDATE courses set unit_price=2750.00 where course_id=104 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 0.000 sec
20 22:30:35 UPDATE courses set unit_price=3500.00 where course_id=105 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 0.015 sec
21 22:31:16 INSERT INTO purchases (purchase_id, learner_id, course_id, quantity, purchase_date) VALUES (1, 1, 101, 1,... 8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0 0.015 sec
22 22:31:52 SELECT COUNT(*) FROM learners LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
Table: courses
Columns:
course_id int PK
course_name varchar(10
category varchar(50
unit_price decimal(10
Query Completed

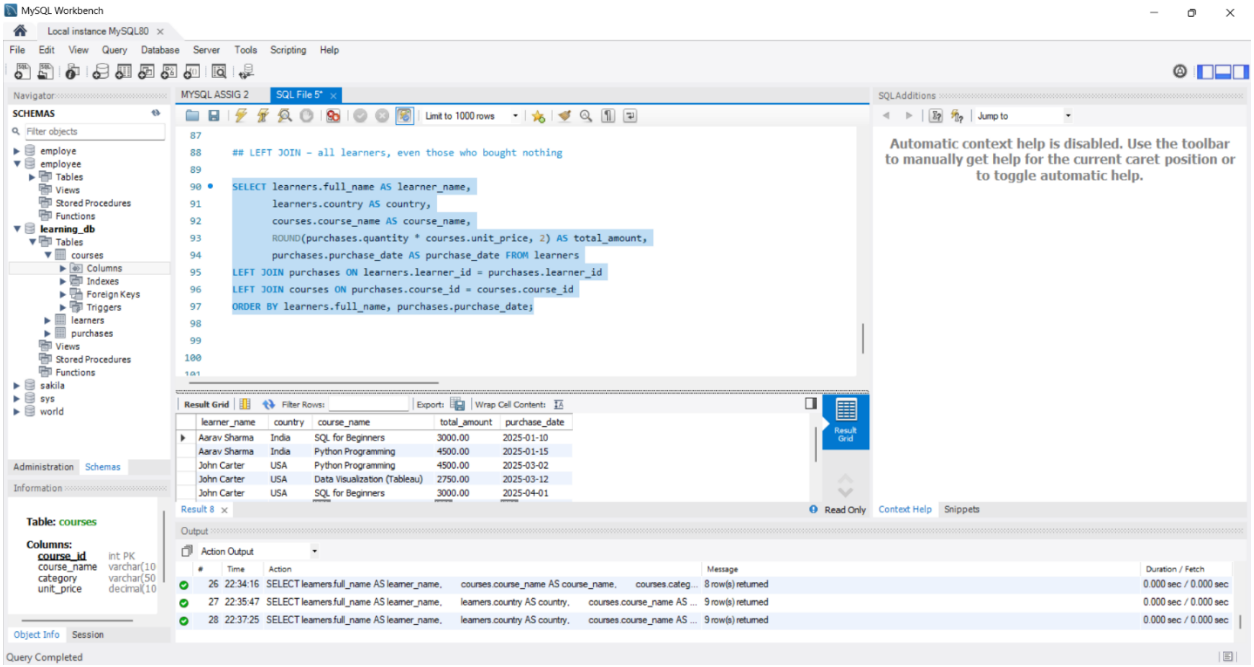


4.Data Exploration Using Joins

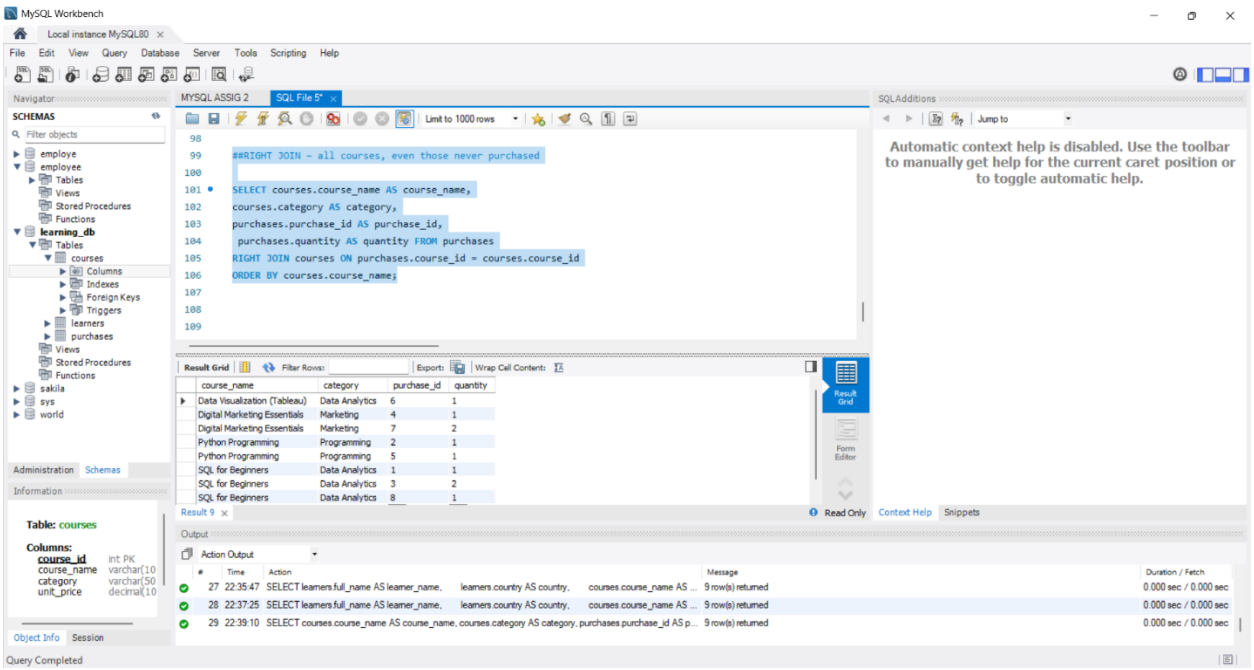
4a. INNER JOIN – each learner's purchase details



4b. LEFT JOIN – all learners, even those who bought nothing

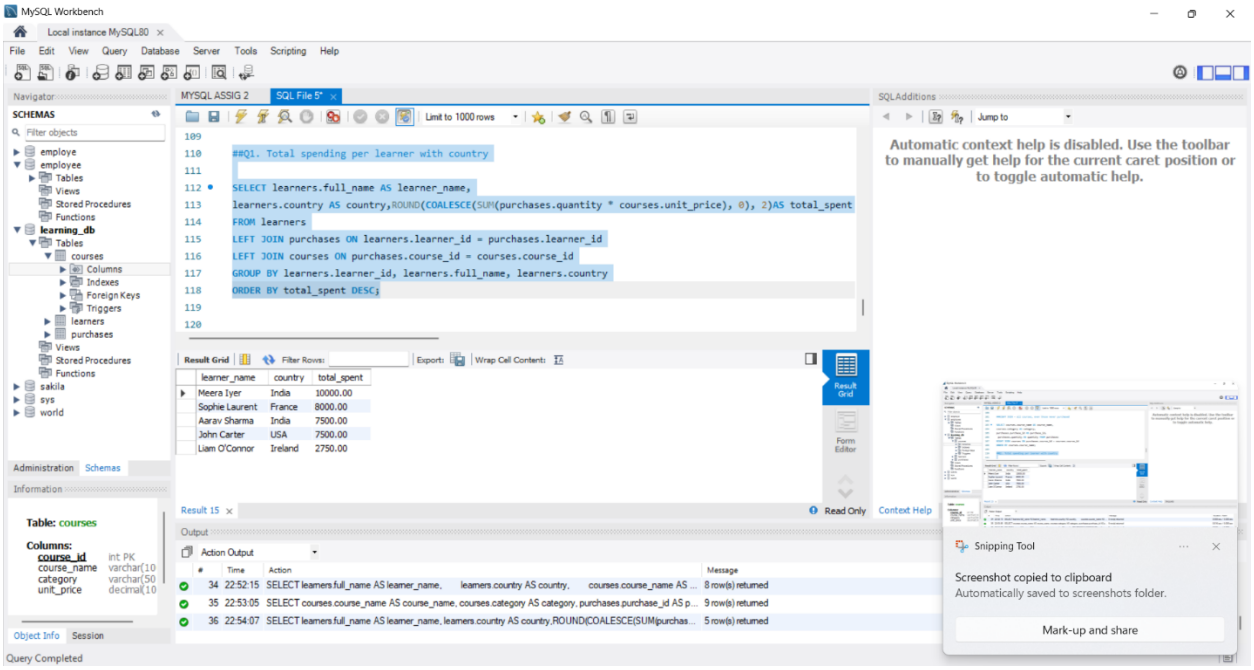


4c. RIGHT JOIN – all courses, even those never purchased

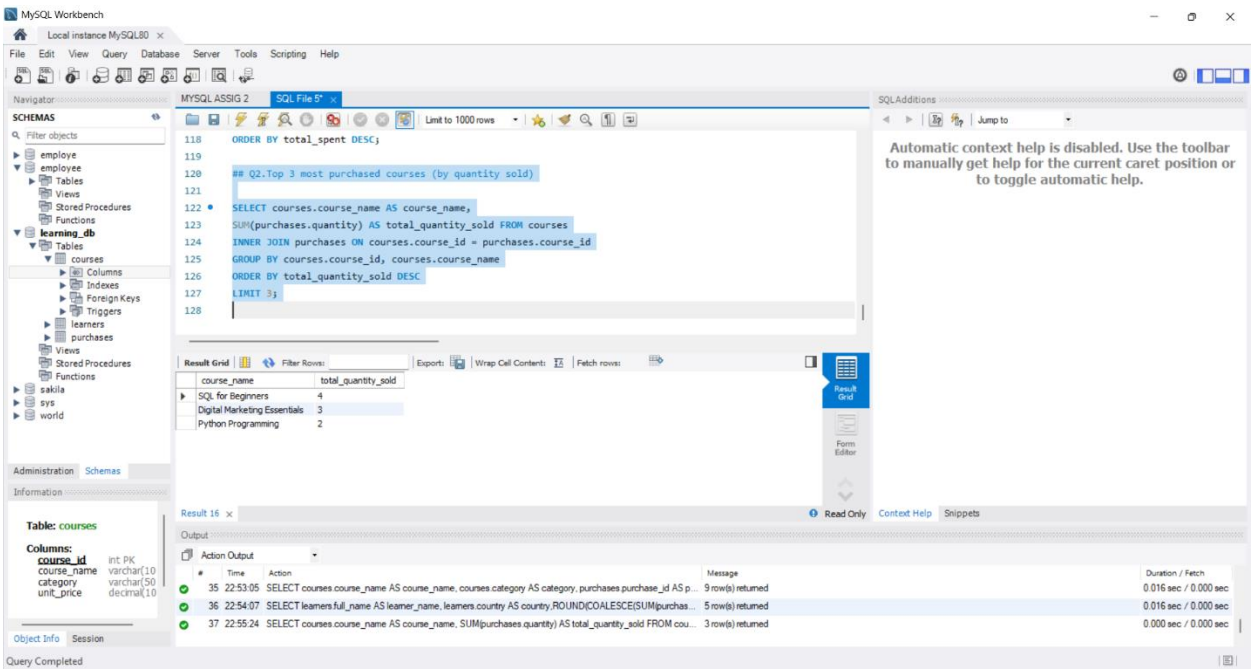


3. Analytical Queries

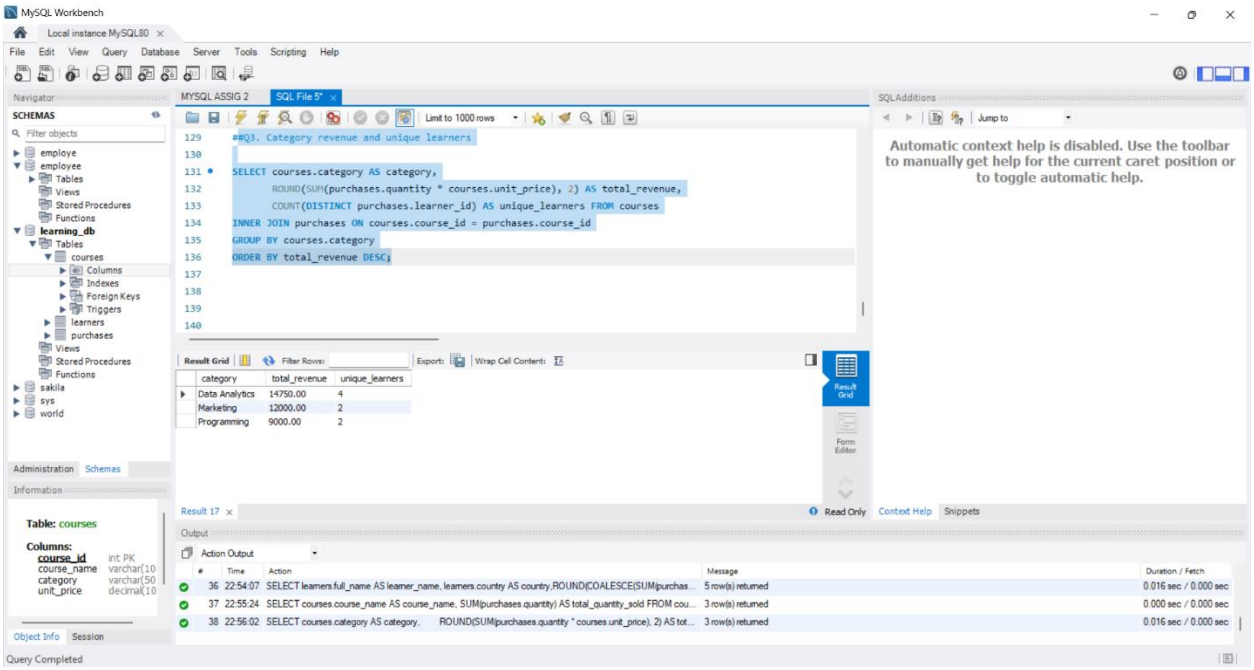
Q1. Total spending per learner with country



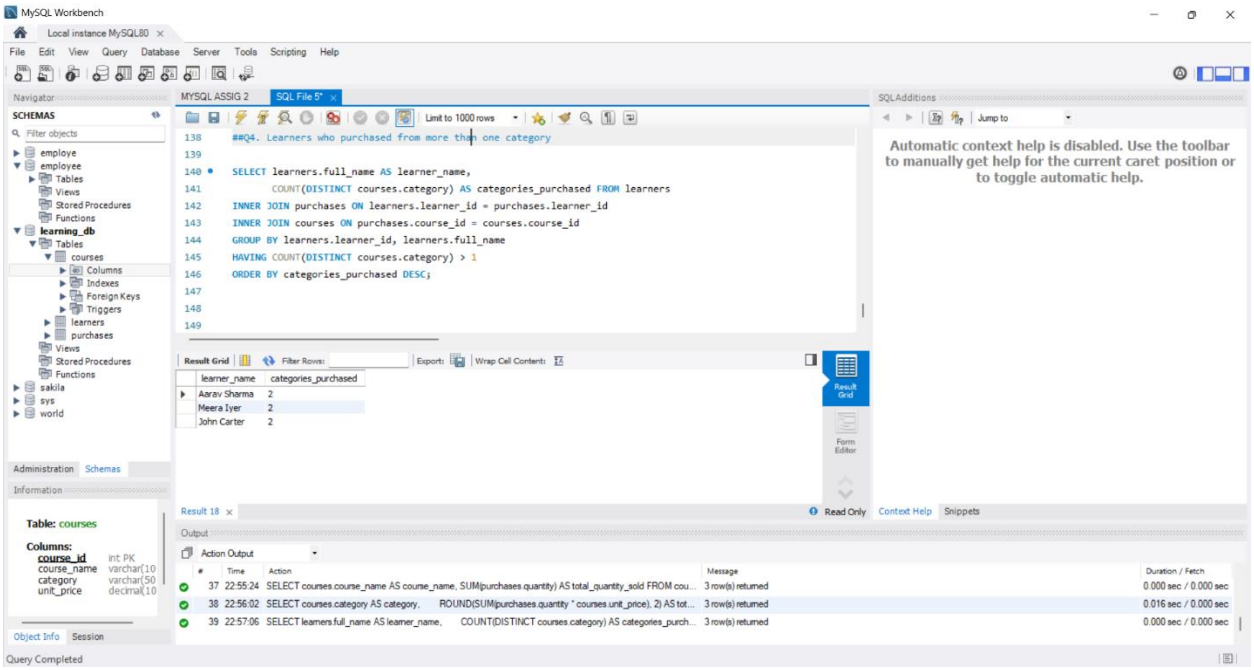
Q2. Top 3 most purchased courses (by quantity sold)



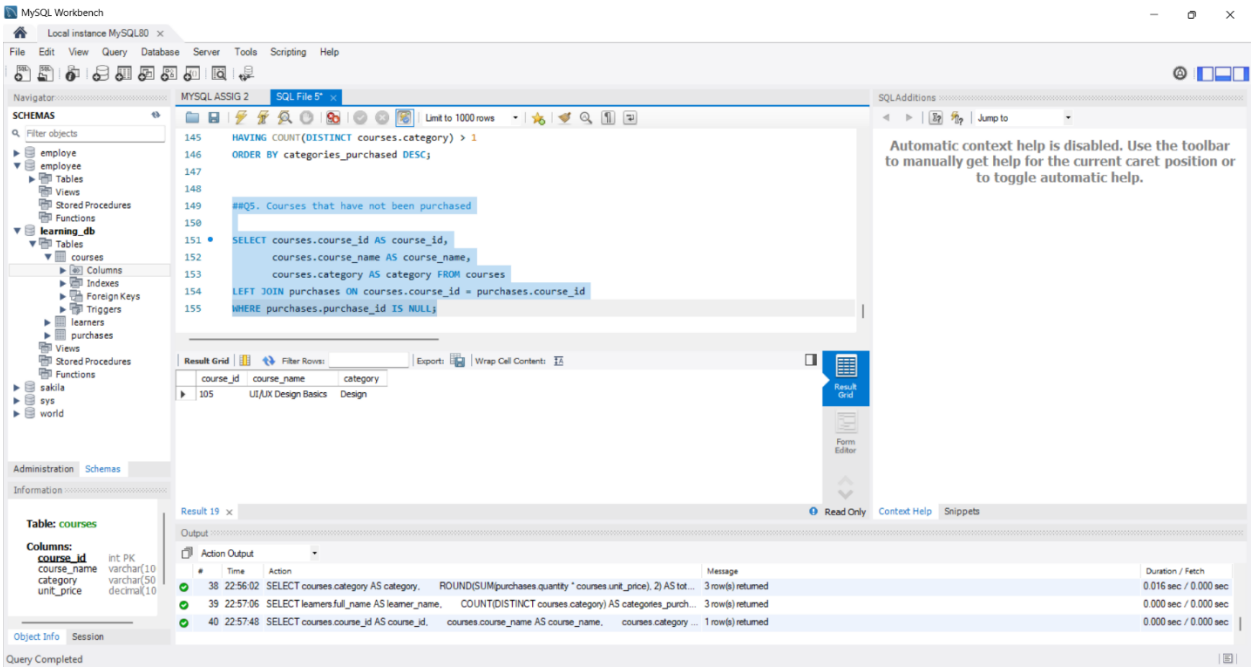
Q3. Category revenue and unique learners



Q4. Learners who purchased from more than one category



Q5. Courses that have not been purchased



Key Insights

• Revenue by category (Q3): Data Analytics generated ₹14,750 (41% of revenue), Marketing ₹12,000 (34%) and Programming ₹9,000 (25%). Data Analytics also reached the most learners (4 of 5).

• Popular courses (Q2): SQL for Beginners sold 4 units, Digital Marketing Essentials 3 units and Python Programming 2 units. The first two each earned ₹12,000.

•Learner spending (Q1): Meera Iyer spent the most (₹10,000), followed by Sophie Laurent (₹8,000), and John Carter and Aarav Sharma (₹7,500 each). Learners from India contributed ₹17,500, about 49% of revenue.

• Cross-category buying (Q4): 3 of 5 learners (Aarav Sharma, Meera Iyer and John Carter) bought from more than one category.

• Unsold course (Q5): UI/UX Design Basics (Design, ₹3,500) had no purchases, making Design the only category with zero sales.

Table 1: Revenue and reach by course category Category Revenue (₹) Share of revenue Unique learners Data Analytics 14,750 41% 4 Marketing 12,000 34% 2 Programming 9,000 25% 2 Total 35,750 100%

Recommendations

• Focus marketing on Data Analytics and Marketing courses, which together bring in 75% of revenue.

• Promote bundles that pair SQL for Beginners with Python Programming or Data Visualization, since 60% of learners already buy across categories

. • Review UI/UX Design Basics (course page, pricing or a launch discount) before spending on broad advertising for it.

• Run localised promotions for India, the largest market, and reward top-spending learners to encourage repeat purchases.

Conclusion

The sample records show ₹35,750 in revenue from 10 course units. Data Analytics leads on both revenue and reach, Marketing is a strong second, and Design needs attention. These findings rest on a small illustrative dataset, so they should be validated against the platform's full transaction data before making business decisions.